

ANALYTICAL REPORT

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Laboratory Job ID: 190-26422-1

Client Project/Site: City of Cadillac Biosolids PFAS

For:

City of Cadillac Utilities
1121 Plett Road
Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

Sue Schafer

Authorized for release by:
8/6/2021 9:28:31 AM

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Results relate only to the items tested and the sample(s) as received by the laboratory.

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Sample Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received
190-26422-1	Biosolids	Solid	07/20/21 13:30	07/21/21 10:46
190-26422-2	Biosolids-Duplicate	Solid	07/20/21 13:30	07/21/21 10:46
190-26422-3	Equipment Blank	Water	07/20/21 13:15	07/21/21 10:46

Case Narrative

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Job ID: 190-26422-1

Laboratory: Eurofins TestAmerica, Michigan

Narrative

Job Narrative 190-26422-1

Comments

No additional comments.

Receipt

The samples were received on 7/21/2021 10:46 AM. Unless otherwise noted below, the samples arrived in good condition, and where required, properly preserved and on ice. The temperature of the cooler at receipt was 4.7° C.

LCMS

Method 537 (modified): The continuing calibration verification (CCVIS) associated with batch 320-510361 recovered above the upper control limit for Perfluorodecanoic acid (PFDA). The samples associated with this CCV were non-detects for the affected analyte; therefore, the data have been reported.

Method 537 (modified): The matrix spike (MS) recoveries for several analytes in preparation batch 320-509488 and analytical batch 320-510044 were outside control limits. Sample matrix interference is suspected because the associated laboratory control sample (LCS) recovery was within acceptance limits.

Method 537 (modified): The "I" qualifier means the transition mass ratio for the indicated analyte was outside of the established ratio limits. The qualitative identification of the analyte has some degree of uncertainty, and the reported value may have some high bias. However, analyst judgement was used to positively identify the analyte.
Biosolids (190-26422-1) and Biosolids-Duplicate (190-26422-2)

Method 537 (modified): Isotope Dilution Analyte (IDA) recovery is above the method recommended limit for the following samples: Biosolids (190-26422-1), Biosolids-Duplicate (190-26422-2), (190-26422-A-1-C MS) and (190-26422-A-1-D MSD). Quantitation by isotope dilution generally precludes any adverse effect on data quality due to elevated IDA recoveries.

Method 537 (modified): The Isotope Dilution Analyte (IDA) recovery associated with the following samples is below the method recommended limit: Biosolids (190-26422-1), Biosolids-Duplicate (190-26422-2) and (190-26422-A-1-C MS). Generally, data quality is not considered affected if the IDA signal-to-noise ratio is greater than 10:1, which is achieved for all IDA in the samples.

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

General Chemistry

No analytical or quality issues were noted, other than those described in the Definitions/Glossary page.

Organic Prep

Method 3535: Insufficient sample volume was available to perform a matrix spike/matrix spike duplicate (MS/MSD) associated with preparation batch 320-510162.

Method: 3535_PFC

Matrix: Water

Method SHAKE: The following samples were yellow after extraction: Biosolids (190-26422-1), Biosolids-Duplicate (190-26422-2), (190-26422-A-1 MS) and (190-26422-A-1 MSD).

preparation batch 320-509488

PFC_IDA

Solid

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Client Sample ID: Biosolids

Lab Sample ID: 190-26422-1

Date Collected: 07/20/21 13:30

Matrix: Solid

Date Received: 07/21/21 10:46

Percent Solids: 3.5

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
F-53B Major	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
F-53B Minor	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
4:2 FTS	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
6:2 FTS	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
8:2 FTS	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
HFPO-DA (GenX)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	8.8	F1	5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	21		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorobutanesulfonic acid (PFBS)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorobutanoic acid (PFBA)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorodecanesulfonic acid (PFDS)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorodecanoic acid (PFDA)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorododecanoic acid (PFDoA)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluoroheptanesulfonic Acid (PFHpS)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluoroheptanoic acid (PFHpA)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorohexanesulfonic acid (PFHxS)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorohexanoic acid (PFHxA)	8.3	I	5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorononanesulfonic acid (PFNS)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorononanoic acid (PFNA)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorooctanesulfonamide (FOSA)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorooctanesulfonic acid (PFOS)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorooctanoic acid (PFOA)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluoropentanesulfonic acid (PFPeS)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluoropentanoic acid (PFPeA)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorotetradecanoic acid (PFTeA)	<5.3	F1	5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluorotridecanoic acid (PFTriA)	<5.3	F1	5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1
Perfluoroundecanoic acid (PFUnA)	<5.3		5.3	ug/Kg	☼	07/23/21 04:42	07/24/21 19:01	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	84		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C3 HFPO-DA	76		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C4 PFBA	67		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C3 PFBS	76		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C2 PFDA	83		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C2 PFDoA	37		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C4 PFHpA	85		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C2 PFHxA	76		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C5 PFNA	87		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C4 PFOA	87		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C4 PFOS	83		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C5 PFPeA	77		25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C2 PFTeDA	24	*5-	25 - 150	07/23/21 04:42	07/24/21 19:01	1
13C2 PFUnA	73		25 - 150	07/23/21 04:42	07/24/21 19:01	1
d5-NEtFOSAA	47		25 - 150	07/23/21 04:42	07/24/21 19:01	1
d3-NMeFOSAA	68		25 - 150	07/23/21 04:42	07/24/21 19:01	1

Eurofins TestAmerica, Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Client Sample ID: Biosolids

Lab Sample ID: 190-26422-1

Date Collected: 07/20/21 13:30

Matrix: Solid

Date Received: 07/21/21 10:46

Percent Solids: 3.5

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
M2-4:2 FTS	155	*5+	25 - 150	07/23/21 04:42	07/24/21 19:01	1
M2-6:2 FTS	196	*5+	25 - 150	07/23/21 04:42	07/24/21 19:01	1
M2-8:2 FTS	205	*5+	25 - 150	07/23/21 04:42	07/24/21 19:01	1
18O2 PFHxS	90		25 - 150	07/23/21 04:42	07/24/21 19:01	1

General Chemistry

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Percent Moisture	96.5		0.1	%			07/23/21 10:03	1
Percent Solids	3.5		0.1	%			07/23/21 10:03	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Client Sample ID: Biosolids-Duplicate

Lab Sample ID: 190-26422-2

Date Collected: 07/20/21 13:30

Matrix: Solid

Date Received: 07/21/21 10:46

Percent Solids: 3.6

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
F-53B Major	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
F-53B Minor	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
4:2 FTS	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
6:2 FTS	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
8:2 FTS	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
HFPO-DA (GenX)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	7.3		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	21		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorobutanesulfonic acid (PFBS)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorobutanoic acid (PFBA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorodecanesulfonic acid (PFDS)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorodecanoic acid (PFDA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorododecanoic acid (PFDoA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluoroheptanesulfonic Acid (PFHpS)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluoroheptanoic acid (PFHpA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorohexanesulfonic acid (PFHxS)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorohexanoic acid (PFHxA)	6.9	I	5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorononanesulfonic acid (PFNS)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorononanoic acid (PFNA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorooctanesulfonamide (FOSA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorooctanesulfonic acid (PFOS)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorooctanoic acid (PFOA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluoropentanesulfonic acid (PFPeS)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluoropentanoic acid (PFPeA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorotetradecanoic acid (PFTeA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluorotridecanoic acid (PFTriA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1
Perfluoroundecanoic acid (PFUnA)	<5.5		5.5	ug/Kg	✱	07/23/21 04:42	07/24/21 19:29	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	75		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C3 HFPO-DA	69		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C4 PFBA	65		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C3 PFBS	69		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C2 PFDA	71		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C2 PFDoA	39		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C4 PFHpA	73		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C2 PFHxA	66		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C5 PFNA	87		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C4 PFOA	86		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C4 PFOS	70		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C5 PFPeA	67		25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C2 PFTeDA	23	*5-	25 - 150	07/23/21 04:42	07/24/21 19:29	1
13C2 PFUnA	66		25 - 150	07/23/21 04:42	07/24/21 19:29	1
d5-NEtFOSAA	42		25 - 150	07/23/21 04:42	07/24/21 19:29	1
d3-NMeFOSAA	65		25 - 150	07/23/21 04:42	07/24/21 19:29	1

Eurofins TestAmerica, Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Client Sample ID: Biosolids-Duplicate

Lab Sample ID: 190-26422-2

Date Collected: 07/20/21 13:30

Matrix: Solid

Date Received: 07/21/21 10:46

Percent Solids: 3.6

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
M2-4:2 FTS	150		25 - 150	07/23/21 04:42	07/24/21 19:29	1
M2-6:2 FTS	187	*5+	25 - 150	07/23/21 04:42	07/24/21 19:29	1
M2-8:2 FTS	156	*5+	25 - 150	07/23/21 04:42	07/24/21 19:29	1
18O2 PFHxS	71		25 - 150	07/23/21 04:42	07/24/21 19:29	1

General Chemistry

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Percent Moisture	96.4		0.1	%			07/23/21 10:03	1
Percent Solids	3.6		0.1	%			07/23/21 10:03	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Client Sample ID: Equipment Blank

Lab Sample ID: 190-26422-3

Date Collected: 07/20/21 13:15

Matrix: Water

Date Received: 07/21/21 10:46

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
F-53B Major	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
F-53B Minor	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
4:2 FTS	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
6:2 FTS	<4.7		4.7	ng/L		07/25/21 19:29	07/26/21 20:47	1
8:2 FTS	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
HFPO-DA (GenX)	<3.7		3.7	ng/L		07/25/21 19:29	07/26/21 20:47	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<4.7		4.7	ng/L		07/25/21 19:29	07/26/21 20:47	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<4.7		4.7	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorobutanesulfonic acid (PFBS)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorobutanoic acid (PFBA)	<4.7		4.7	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorodecanesulfonic acid (PFDS)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorodecanoic acid (PFDA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorododecanoic acid (PFDoA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluoroheptanoic acid (PFHpA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorohexanesulfonic acid (PFHxS)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorohexanoic acid (PFHxA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorononanesulfonic acid (PFNS)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorononanoic acid (PFNA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorooctanesulfonamide (FOSA)	2.1		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorooctanesulfonic acid (PFOS)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorooctanoic acid (PFOA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluoropentanesulfonic acid (PFPeS)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluoropentanoic acid (PFPeA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorotetradecanoic acid (PFTeA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluorotridecanoic acid (PFTrIA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1
Perfluoroundecanoic acid (PFUnA)	<1.9		1.9	ng/L		07/25/21 19:29	07/26/21 20:47	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	87		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C3 HFPO-DA	90		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C4 PFBA	93		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C3 PFBS	89		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C2 PFDA	88		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C2 PFDoA	82		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C4 PFHpA	94		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C2 PFHxA	94		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C5 PFNA	101		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C4 PFOA	94		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C4 PFOS	95		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C5 PFPeA	92		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C2 PFTeDA	74		25 - 150	07/25/21 19:29	07/26/21 20:47	1
13C2 PFUnA	91		25 - 150	07/25/21 19:29	07/26/21 20:47	1
d5-NEtFOSAA	89		25 - 150	07/25/21 19:29	07/26/21 20:47	1
d3-NMeFOSAA	87		25 - 150	07/25/21 19:29	07/26/21 20:47	1

Eurofins TestAmerica, Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Client Sample ID: Equipment Blank

Lab Sample ID: 190-26422-3

Date Collected: 07/20/21 13:15

Matrix: Water

Date Received: 07/21/21 10:46

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
M2-4:2 FTS	127		25 - 150	07/25/21 19:29	07/26/21 20:47	1
M2-6:2 FTS	140		25 - 150	07/25/21 19:29	07/26/21 20:47	1
M2-8:2 FTS	131		25 - 150	07/25/21 19:29	07/26/21 20:47	1
18O2 PFHxS	97		25 - 150	07/25/21 19:29	07/26/21 20:47	1

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Lab Sample ID: MB 320-509488/1-A

Matrix: Solid

Analysis Batch: 510044

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 509488

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
F-53B Major	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
F-53B Minor	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
4:2 FTS	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
6:2 FTS	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
8:2 FTS	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
HFPO-DA (GenX)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorobutanesulfonic acid (PFBS)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorobutanoic acid (PFBA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorodecanesulfonic acid (PFDS)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorodecanoic acid (PFDA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorododecanoic acid (PFDoA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluoroheptanesulfonic Acid (PFHpS)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluoroheptanoic acid (PFHpA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorohexanesulfonic acid (PFHxS)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorohexanoic acid (PFHxA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorononanesulfonic acid (PFNS)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorononanoic acid (PFNA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorooctanesulfonamide (FOSA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorooctanesulfonic acid (PFOS)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorooctanoic acid (PFOA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluoropentanesulfonic acid (PFPeS)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluoropentanoic acid (PFPeA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorotetradecanoic acid (PFTeA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluorotridecanoic acid (PFTriA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1
Perfluoroundecanoic acid (PFUnA)	<0.20		0.20	ug/Kg		07/23/21 04:42	07/24/21 18:34	1

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	89		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C3 HFPO-DA	87		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C4 PFBA	85		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C3 PFBS	81		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C2 PFDA	84		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C2 PFDoA	83		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C4 PFHpA	100		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C2 PFHxA	83		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C5 PFNA	93		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C4 PFOA	88		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C4 PFOS	98		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C5 PFPeA	88		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C2 PFTeA	81		25 - 150	07/23/21 04:42	07/24/21 18:34	1
13C2 PFUnA	89		25 - 150	07/23/21 04:42	07/24/21 18:34	1
d5-NEtFOSAA	88		25 - 150	07/23/21 04:42	07/24/21 18:34	1

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: MB 320-509488/1-A

Matrix: Solid

Analysis Batch: 510044

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 509488

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
d3-NMeFOSAA	81		25 - 150	07/23/21 04:42	07/24/21 18:34	1
M2-4:2 FTS	72		25 - 150	07/23/21 04:42	07/24/21 18:34	1
M2-6:2 FTS	77		25 - 150	07/23/21 04:42	07/24/21 18:34	1
M2-8:2 FTS	84		25 - 150	07/23/21 04:42	07/24/21 18:34	1
18O2 PFHxS	95		25 - 150	07/23/21 04:42	07/24/21 18:34	1

Lab Sample ID: LCS 320-509488/2-A

Matrix: Solid

Analysis Batch: 510044

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 509488

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	1.88	2.03		ug/Kg		108	79 - 139
F-53B Major	1.86	1.91		ug/Kg		103	74 - 134
F-53B Minor	1.88	1.70		ug/Kg		90	66 - 136
4:2 FTS	1.87	1.99		ug/Kg		106	68 - 143
6:2 FTS	1.90	2.20		ug/Kg		116	73 - 139
8:2 FTS	1.92	2.17		ug/Kg		113	75 - 135
HFPO-DA (GenX)	2.00	2.36		ug/Kg		118	53 - 158
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	2.00	2.09		ug/Kg		105	72 - 132
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	2.00	2.27		ug/Kg		113	72 - 132
Perfluorobutanesulfonic acid (PFBS)	1.77	2.04		ug/Kg		116	69 - 129
Perfluorobutanoic acid (PFBA)	2.00	2.23		ug/Kg		112	76 - 136
Perfluorodecanesulfonic acid (PFDS)	1.93	2.01		ug/Kg		104	71 - 131
Perfluorodecanoic acid (PFDA)	2.00	2.21		ug/Kg		110	72 - 132
Perfluorododecanoic acid (PFDoA)	2.00	2.23		ug/Kg		112	71 - 131
Perfluoroheptanesulfonic Acid (PFHpS)	1.90	2.18		ug/Kg		114	76 - 136
Perfluoroheptanoic acid (PFHpA)	2.00	2.23		ug/Kg		111	71 - 131
Perfluorohexanesulfonic acid (PFHxS)	1.82	2.02		ug/Kg		111	62 - 122
Perfluorohexanoic acid (PFHxA)	2.00	2.24		ug/Kg		112	71 - 131
Perfluorononanesulfonic acid (PFNS)	1.92	1.87		ug/Kg		98	72 - 132
Perfluorononanoic acid (PFNA)	2.00	2.26		ug/Kg		113	73 - 133
Perfluorooctanesulfonamide (FOSA)	2.00	2.12		ug/Kg		106	77 - 137
Perfluorooctanesulfonic acid (PFOS)	1.86	2.06		ug/Kg		111	68 - 141
Perfluorooctanoic acid (PFOA)	2.00	2.23		ug/Kg		111	72 - 132
Perfluoropentanesulfonic acid (PFPeS)	1.88	2.18		ug/Kg		116	66 - 126
Perfluoropentanoic acid (PFPeA)	2.00	2.30		ug/Kg		115	69 - 129
Perfluorotetradecanoic acid (PFTeA)	2.00	2.19		ug/Kg		110	67 - 127
Perfluorotridecanoic acid (PFTriA)	2.00	2.48		ug/Kg		124	71 - 131

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-509488/2-A

Matrix: Solid

Analysis Batch: 510044

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 509488

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Perfluoroundecanoic acid (PFUnA)	2.00	2.25		ug/Kg		113	66 - 126
	LCS	LCS					
Isotope Dilution	%Recovery	Qualifier	Limits				
13C8 FOSA	98		25 - 150				
13C3 HFPO-DA	79		25 - 150				
13C4 PFBA	90		25 - 150				
13C3 PFBS	83		25 - 150				
13C2 PFDA	83		25 - 150				
13C2 PFDoA	77		25 - 150				
13C4 PFHpA	100		25 - 150				
13C2 PFHxA	84		25 - 150				
13C5 PFNA	88		25 - 150				
13C4 PFOA	85		25 - 150				
13C4 PFOS	92		25 - 150				
13C5 PFPeA	89		25 - 150				
13C2 PFTeDA	79		25 - 150				
13C2 PFUnA	81		25 - 150				
d5-NEtFOSAA	80		25 - 150				
d3-NMeFOSAA	78		25 - 150				
M2-4:2 FTS	63		25 - 150				
M2-6:2 FTS	66		25 - 150				
M2-8:2 FTS	76		25 - 150				
18O2 PFHxS	92		25 - 150				

Lab Sample ID: 190-26422-1 MS

Matrix: Solid

Analysis Batch: 510044

Client Sample ID: Biosolids

Prep Type: Total/NA

Prep Batch: 509488

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<5.3		49.2	62.4		ug/Kg	✱	127	79 - 139
F-53B Major	<5.3		48.7	58.7		ug/Kg	✱	120	74 - 134
F-53B Minor	<5.3		49.2	41.3		ug/Kg	✱	84	66 - 136
4:2 FTS	<5.3		48.8	49.3		ug/Kg	✱	101	68 - 143
6:2 FTS	<5.3		49.5	58.9		ug/Kg	✱	119	73 - 139
8:2 FTS	<5.3		50.1	49.9		ug/Kg	✱	100	75 - 135
HFPO-DA (GenX)	<5.3		52.3	57.9		ug/Kg	✱	111	53 - 158
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	8.8	F1	52.3	80.5	F1	ug/Kg	✱	137	72 - 132
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	21		52.3	78.8		ug/Kg	✱	110	72 - 132
Perfluorobutanesulfonic acid (PFBS)	<5.3		46.2	59.8		ug/Kg	✱	127	69 - 129
Perfluorobutanoic acid (PFBA)	<5.3		52.3	58.7		ug/Kg	✱	112	76 - 136
Perfluorodecanesulfonic acid (PFDS)	<5.3		50.4	50.8		ug/Kg	✱	101	71 - 131
Perfluorodecanoic acid (PFDA)	<5.3		52.3	56.4		ug/Kg	✱	105	72 - 132
Perfluorododecanoic acid (PFDoA)	<5.3		52.3	54.2		ug/Kg	✱	104	71 - 131

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: 190-26422-1 MS

Matrix: Solid

Analysis Batch: 510044

Client Sample ID: Biosolids

Prep Type: Total/NA

Prep Batch: 509488

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
Perfluoroheptanesulfonic Acid (PFHpS)	<5.3		49.8	64.9		ug/Kg	⊛	130	76 - 136
Perfluoroheptanoic acid (PFHpA)	<5.3		52.3	54.2		ug/Kg	⊛	104	71 - 131
Perfluorohexanesulfonic acid (PFHxS)	<5.3		47.6	56.6		ug/Kg	⊛	119	62 - 122
Perfluorohexanoic acid (PFHxA)	8.3	I	52.3	68.2		ug/Kg	⊛	115	71 - 131
Perfluorononanesulfonic acid (PFNS)	<5.3		50.2	60.7		ug/Kg	⊛	121	72 - 132
Perfluorononanoic acid (PFNA)	<5.3		52.3	54.7		ug/Kg	⊛	105	73 - 133
Perfluorooctanesulfonamide (FOSA)	<5.3		52.3	57.3		ug/Kg	⊛	107	77 - 137
Perfluorooctanesulfonic acid (PFOS)	<5.3		48.5	59.1		ug/Kg	⊛	114	68 - 141
Perfluorooctanoic acid (PFOA)	<5.3		52.3	61.4		ug/Kg	⊛	117	72 - 132
Perfluoropentanesulfonic acid (PFPeS)	<5.3		49.0	58.4		ug/Kg	⊛	119	66 - 126
Perfluoropentanoic acid (PFPeA)	<5.3		52.3	64.2		ug/Kg	⊛	123	69 - 129
Perfluorotetradecanoic acid (PFTeA)	<5.3	F1	52.3	70.7	F1	ug/Kg	⊛	135	67 - 127
Perfluorotridecanoic acid (PFTriA)	<5.3	F1	52.3	30.1	F1	ug/Kg	⊛	58	71 - 131
Perfluoroundecanoic acid (PFUnA)	<5.3		52.3	57.5		ug/Kg	⊛	110	66 - 126
Isotope Dilution	MS %Recovery	MS Qualifier	Limits						
13C8 FOSA	80		25 - 150						
13C3 HFPO-DA	74		25 - 150						
13C4 PFBA	61		25 - 150						
13C3 PFBS	71		25 - 150						
13C2 PFDA	72		25 - 150						
13C2 PFDoA	39		25 - 150						
13C4 PFHpA	87		25 - 150						
13C2 PFHxA	72		25 - 150						
13C5 PFNA	85		25 - 150						
13C4 PFOA	87		25 - 150						
13C4 PFOS	70		25 - 150						
13C5 PFPeA	68		25 - 150						
13C2 PFTeDA	23	*5-	25 - 150						
13C2 PFUnA	66		25 - 150						
d5-NEtFOSAA	41		25 - 150						
d3-NMeFOSAA	72		25 - 150						
M2-4:2 FTS	150		25 - 150						
M2-6:2 FTS	179	*5+	25 - 150						
M2-8:2 FTS	191	*5+	25 - 150						
18O2 PFHxS	81		25 - 150						

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: 190-26422-1 MSD

Matrix: Solid

Analysis Batch: 510044

Client Sample ID: Biosolids

Prep Type: Total/NA

Prep Batch: 509488

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<5.3		51.7	68.3		ug/Kg	✱	132	79 - 139	9	30
F-53B Major	<5.3		51.1	61.6		ug/Kg	✱	121	74 - 134	5	30
F-53B Minor	<5.3		51.7	47.4		ug/Kg	✱	92	66 - 136	14	30
4:2 FTS	<5.3		51.2	55.1		ug/Kg	✱	108	68 - 143	11	30
6:2 FTS	<5.3		52.0	57.8		ug/Kg	✱	111	73 - 139	2	30
8:2 FTS	<5.3		52.5	58.4		ug/Kg	✱	111	75 - 135	16	30
HFPO-DA (GenX)	<5.3		54.8	60.5		ug/Kg	✱	110	53 - 158	4	30
N-ethylperfluorooctanesulfonamidoacetic acid (NETFOSAA)	8.8	F1	54.8	74.4		ug/Kg	✱	120	72 - 132	8	30
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	21		54.8	87.0		ug/Kg	✱	120	72 - 132	10	30
Perfluorobutanesulfonic acid (PFBS)	<5.3		48.5	57.7		ug/Kg	✱	117	69 - 129	3	30
Perfluorobutanoic acid (PFBA)	<5.3		54.8	61.4		ug/Kg	✱	112	76 - 136	4	30
Perfluorodecanesulfonic acid (PFDS)	<5.3		52.9	55.8		ug/Kg	✱	106	71 - 131	9	30
Perfluorodecanoic acid (PFDA)	<5.3		54.8	61.1		ug/Kg	✱	109	72 - 132	8	30
Perfluorododecanoic acid (PFDoA)	<5.3		54.8	69.4		ug/Kg	✱	127	71 - 131	25	30
Perfluoroheptanesulfonic Acid (PFHpS)	<5.3		52.2	68.3		ug/Kg	✱	131	76 - 136	5	30
Perfluoroheptanoic acid (PFHpA)	<5.3		54.8	58.4		ug/Kg	✱	106	71 - 131	7	30
Perfluorohexanesulfonic acid (PFHxS)	<5.3		49.9	55.0		ug/Kg	✱	110	62 - 122	3	30
Perfluorohexanoic acid (PFHxA)	8.3	I	54.8	67.8		ug/Kg	✱	108	71 - 131	1	30
Perfluorononanesulfonic acid (PFNS)	<5.3		52.7	56.9		ug/Kg	✱	108	72 - 132	6	30
Perfluorononanoic acid (PFNA)	<5.3		54.8	58.4		ug/Kg	✱	107	73 - 133	7	30
Perfluorooctanesulfonamide (FOSA)	<5.3		54.8	55.4		ug/Kg	✱	99	77 - 137	3	30
Perfluorooctanesulfonic acid (PFOS)	<5.3		50.9	59.8		ug/Kg	✱	110	68 - 141	1	30
Perfluorooctanoic acid (PFOA)	<5.3		54.8	68.3		ug/Kg	✱	125	72 - 132	11	30
Perfluoropentanesulfonic acid (PFPeS)	<5.3		51.4	61.4		ug/Kg	✱	119	66 - 126	5	30
Perfluoropentanoic acid (PFPeA)	<5.3		54.8	68.7		ug/Kg	✱	125	69 - 129	7	30
Perfluorotetradecanoic acid (PFTeA)	<5.3	F1	54.8	69.1		ug/Kg	✱	126	67 - 127	2	30
Perfluorotridecanoic acid (PFTriA)	<5.3	F1	54.8	38.8		ug/Kg	✱	71	71 - 131	25	30
Perfluoroundecanoic acid (PFUnA)	<5.3		54.8	62.2		ug/Kg	✱	113	66 - 126	8	30

Isotope Dilution	MSD %Recovery	MSD Qualifier	Limits
13C8 FOSA	80		25 - 150
13C3 HFPO-DA	73		25 - 150
13C4 PFBA	71		25 - 150
13C3 PFBS	78		25 - 150
13C2 PFDA	79		25 - 150
13C2 PFDoA	34		25 - 150
13C4 PFHpA	82		25 - 150

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: 190-26422-1 MSD

Matrix: Solid

Analysis Batch: 510044

Client Sample ID: Biosolids

Prep Type: Total/NA

Prep Batch: 509488

Isotope Dilution	MSD		Limits
	%Recovery	Qualifier	
13C2 PFHxA	74		25 - 150
13C5 PFNA	86		25 - 150
13C4 PFOA	82		25 - 150
13C4 PFOS	72		25 - 150
13C5 PFPeA	73		25 - 150
13C2 PFTeDA	26		25 - 150
13C2 PFUnA	68		25 - 150
d5-NEtFOSAA	46		25 - 150
d3-NMeFOSAA	73		25 - 150
M2-4:2 FTS	133		25 - 150
M2-6:2 FTS	186	*5+	25 - 150
M2-8:2 FTS	186	*5+	25 - 150
18O2 PFHxS	84		25 - 150

Lab Sample ID: MB 320-510162/1-A

Matrix: Water

Analysis Batch: 510387

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 510162

Analyte	MB		RL	Unit	D	Prepared	Analyzed	Dil Fac
	Result	Qualifier						
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
F-53B Major	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
F-53B Minor	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
4:2 FTS	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
6:2 FTS	<5.0		5.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
8:2 FTS	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
HFPO-DA (GenX)	<4.0		4.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<5.0		5.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<5.0		5.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorobutanesulfonic acid (PFBS)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorobutanoic acid (PFBA)	<5.0		5.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorodecanesulfonic acid (PFDS)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorodecanoic acid (PFDA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorododecanoic acid (PFDoA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluoroheptanoic acid (PFHpA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorohexanesulfonic acid (PFHxS)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorohexanoic acid (PFHxA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorononanesulfonic acid (PFNS)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorononanoic acid (PFNA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorooctanesulfonamide (FOSA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorooctanesulfonic acid (PFOS)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorooctanoic acid (PFOA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluoropentanesulfonic acid (PFPeS)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluoropentanoic acid (PFPeA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluorotetradecanoic acid (PFTeA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: MB 320-510162/1-A

Matrix: Water

Analysis Batch: 510387

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 510162

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Perfluorotridecanoic acid (PFTriA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Perfluoroundecanoic acid (PFUnA)	<2.0		2.0	ng/L		07/25/21 19:29	07/26/21 19:52	1
Isotope Dilution	MB %Recovery	MB Qualifier	Limits			Prepared	Analyzed	Dil Fac
13C8 FOSA	90		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C3 HFPO-DA	83		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C4 PFBA	87		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C3 PFBS	80		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C2 PFDA	84		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C2 PFDoA	79		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C4 PFHpA	87		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C2 PFHxA	86		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C5 PFNA	86		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C4 PFOA	91		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C4 PFOS	89		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C5 PFPeA	86		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C2 PFTeDA	77		25 - 150			07/25/21 19:29	07/26/21 19:52	1
13C2 PFUnA	94		25 - 150			07/25/21 19:29	07/26/21 19:52	1
d5-NEtFOSAA	86		25 - 150			07/25/21 19:29	07/26/21 19:52	1
d3-NMeFOSAA	85		25 - 150			07/25/21 19:29	07/26/21 19:52	1
M2-4:2 FTS	112		25 - 150			07/25/21 19:29	07/26/21 19:52	1
M2-6:2 FTS	117		25 - 150			07/25/21 19:29	07/26/21 19:52	1
M2-8:2 FTS	108		25 - 150			07/25/21 19:29	07/26/21 19:52	1
18O2 PFHxS	86		25 - 150			07/25/21 19:29	07/26/21 19:52	1

Lab Sample ID: LCS 320-510162/2-A

Matrix: Water

Analysis Batch: 510387

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 510162

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	Limits
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	37.7	37.1		ng/L		99	79 - 139
F-53B Major	37.3	35.1		ng/L		94	75 - 135
F-53B Minor	37.7	35.1		ng/L		93	54 - 114
4:2 FTS	37.4	38.4		ng/L		103	79 - 139
6:2 FTS	37.9	37.1		ng/L		98	59 - 175
8:2 FTS	38.3	40.7		ng/L		106	75 - 135
HFPO-DA (GenX)	40.0	41.4		ng/L		104	51 - 173
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	37.7		ng/L		94	76 - 136
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	37.9		ng/L		95	76 - 136
Perfluorobutanesulfonic acid (PFBS)	35.4	36.2		ng/L		102	67 - 127
Perfluorobutanoic acid (PFBA)	40.0	41.0		ng/L		103	76 - 136
Perfluorodecanesulfonic acid (PFDS)	38.6	35.2		ng/L		91	71 - 131
Perfluorodecanoic acid (PFDA)	40.0	41.8		ng/L		104	76 - 136
Perfluorododecanoic acid (PFDoA)	40.0	42.2		ng/L		105	71 - 131

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-510162/2-A

Matrix: Water

Analysis Batch: 510387

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 510162

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	37.9		ng/L		99	76 - 136
Perfluoroheptanoic acid (PFHpA)	40.0	39.1		ng/L		98	72 - 132
Perfluorohexanesulfonic acid (PFHxS)	36.4	33.7		ng/L		92	59 - 119
Perfluorohexanoic acid (PFHxA)	40.0	38.5		ng/L		96	73 - 133
Perfluorononanesulfonic acid (PFNS)	38.4	35.9		ng/L		93	75 - 135
Perfluorononanoic acid (PFNA)	40.0	41.5		ng/L		104	75 - 135
Perfluorooctanesulfonamide (FOSA)	40.0	38.0		ng/L		95	73 - 133
Perfluorooctanesulfonic acid (PFOS)	37.1	37.9		ng/L		102	70 - 130
Perfluorooctanoic acid (PFOA)	40.0	42.4		ng/L		106	70 - 130
Perfluoropentanesulfonic acid (PFPeS)	37.5	37.7		ng/L		100	66 - 126
Perfluoropentanoic acid (PFPeA)	40.0	38.6		ng/L		96	71 - 131
Perfluorotetradecanoic acid (PFTeA)	40.0	45.7		ng/L		114	70 - 130
Perfluorotridecanoic acid (PFTriA)	40.0	36.6		ng/L		92	71 - 131
Perfluoroundecanoic acid (PFUnA)	40.0	43.6		ng/L		109	68 - 128

Isotope Dilution	LCS %Recovery	LCS Qualifier	Limits
13C8 FOSA	88		25 - 150
13C3 HFPO-DA	86		25 - 150
13C4 PFBA	88		25 - 150
13C3 PFBS	82		25 - 150
13C2 PFDA	85		25 - 150
13C2 PFDoA	82		25 - 150
13C4 PFHpA	92		25 - 150
13C2 PFHxA	88		25 - 150
13C5 PFNA	91		25 - 150
13C4 PFOA	89		25 - 150
13C4 PFOS	92		25 - 150
13C5 PFPeA	86		25 - 150
13C2 PFTeDA	80		25 - 150
13C2 PFUnA	86		25 - 150
d5-NEtFOSAA	85		25 - 150
d3-NMeFOSAA	83		25 - 150
M2-4:2 FTS	106		25 - 150
M2-6:2 FTS	116		25 - 150
M2-8:2 FTS	120		25 - 150
18O2 PFHxS	93		25 - 150

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCSD 320-510162/3-A

Matrix: Water

Analysis Batch: 510387

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

Prep Batch: 510162

Analyte	Spike Added	LCSD Result	LCSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	37.7	37.1		ng/L		98	79 - 139	0	30
F-53B Major	37.3	37.9		ng/L		102	75 - 135	8	30
F-53B Minor	37.7	37.4		ng/L		99	54 - 114	6	30
4:2 FTS	37.4	38.0		ng/L		102	79 - 139	1	30
6:2 FTS	37.9	38.0		ng/L		100	59 - 175	2	30
8:2 FTS	38.3	45.6		ng/L		119	75 - 135	11	30
HFPO-DA (GenX)	40.0	41.6		ng/L		104	51 - 173	0	30
N-ethylperfluorooctanesulfonamidoacetic acid (NETFOSAA)	40.0	40.7		ng/L		102	76 - 136	8	30
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	37.2		ng/L		93	76 - 136	2	30
Perfluorobutanesulfonic acid (PFBS)	35.4	39.8		ng/L		112	67 - 127	9	30
Perfluorobutanoic acid (PFBA)	40.0	40.9		ng/L		102	76 - 136	0	30
Perfluorodecanesulfonic acid (PFDS)	38.6	37.5		ng/L		97	71 - 131	6	30
Perfluorodecanoic acid (PFDA)	40.0	47.0		ng/L		117	76 - 136	12	30
Perfluorododecanoic acid (PFDoA)	40.0	41.8		ng/L		105	71 - 131	1	30
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	39.2		ng/L		103	76 - 136	4	30
Perfluoroheptanoic acid (PFHpA)	40.0	40.9		ng/L		102	72 - 132	4	30
Perfluorohexanesulfonic acid (PFHxS)	36.4	36.3		ng/L		100	59 - 119	7	30
Perfluorohexanoic acid (PFHxA)	40.0	40.6		ng/L		101	73 - 133	5	30
Perfluorononanesulfonic acid (PFNS)	38.4	38.2		ng/L		100	75 - 135	6	30
Perfluorononanoic acid (PFNA)	40.0	45.0		ng/L		112	75 - 135	8	30
Perfluorooctanesulfonamide (FOSA)	40.0	40.0		ng/L		100	73 - 133	5	30
Perfluorooctanesulfonic acid (PFOS)	37.1	39.2		ng/L		106	70 - 130	3	30
Perfluorooctanoic acid (PFOA)	40.0	42.8		ng/L		107	70 - 130	1	30
Perfluoropentanesulfonic acid (PFPeS)	37.5	41.5		ng/L		111	66 - 126	10	30
Perfluoropentanoic acid (PFPeA)	40.0	38.6		ng/L		97	71 - 131	0	30
Perfluorotetradecanoic acid (PFTeA)	40.0	46.5		ng/L		116	70 - 130	2	30
Perfluorotridecanoic acid (PFTriA)	40.0	40.1		ng/L		100	71 - 131	9	30
Perfluoroundecanoic acid (PFUnA)	40.0	40.8		ng/L		102	68 - 128	7	30

Isotope Dilution	LCSD %Recovery	LCSD Qualifier	Limits
13C8 FOSA	89		25 - 150
13C3 HFPO-DA	85		25 - 150
13C4 PFBA	87		25 - 150
13C3 PFBS	79		25 - 150
13C2 PFDA	80		25 - 150
13C2 PFDoA	87		25 - 150
13C4 PFHpA	90		25 - 150

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCSD 320-510162/3-A

Matrix: Water

Analysis Batch: 510387

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

Prep Batch: 510162

Isotope Dilution	LCSD		Limits
	%Recovery	Qualifier	
13C2 PFHxA	85		25 - 150
13C5 PFNA	88		25 - 150
13C4 PFOA	91		25 - 150
13C4 PFOS	91		25 - 150
13C5 PFPeA	88		25 - 150
13C2 PFTeDA	78		25 - 150
13C2 PFUnA	92		25 - 150
d5-NEtFOSAA	83		25 - 150
d3-NMeFOSAA	87		25 - 150
M2-4:2 FTS	111		25 - 150
M2-6:2 FTS	111		25 - 150
M2-8:2 FTS	102		25 - 150
18O2 PFHxS	91		25 - 150

Isotope Dilution Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Solid

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOSA (25-150)	HFPODA (25-150)	PFBA (25-150)	C3PFBS (25-150)	PFDA (25-150)	PFDoA (25-150)	C4PFHA (25-150)	PFHxA (25-150)
190-26422-1	Biosolids	84	76	67	76	83	37	85	76
190-26422-1 MS	Biosolids	80	74	61	71	72	39	87	72
190-26422-1 MSD	Biosolids	80	73	71	78	79	34	82	74
190-26422-2	Biosolids-Duplicate	75	69	65	69	71	39	73	66
LCS 320-509488/2-A	Lab Control Sample	98	79	90	83	83	77	100	84
MB 320-509488/1-A	Method Blank	89	87	85	81	84	83	100	83

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFNA (25-150)	PFOA (25-150)	PFOS (25-150)	PFPeA (25-150)	PFTDA (25-150)	PFUnA (25-150)	d5NEFOS (25-150)	d3NMFOS (25-150)
190-26422-1	Biosolids	87	87	83	77	24 *5-	73	47	68
190-26422-1 MS	Biosolids	85	87	70	68	23 *5-	66	41	72
190-26422-1 MSD	Biosolids	86	82	72	73	26	68	46	73
190-26422-2	Biosolids-Duplicate	87	86	70	67	23 *5-	66	42	65
LCS 320-509488/2-A	Lab Control Sample	88	85	92	89	79	81	80	78
MB 320-509488/1-A	Method Blank	93	88	98	88	81	89	88	81

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M242FTS (25-150)	M262FTS (25-150)	M282FTS (25-150)	PFHxS (25-150)
190-26422-1	Biosolids	155 *5+	196 *5+	205 *5+	90
190-26422-1 MS	Biosolids	150	179 *5+	191 *5+	81
190-26422-1 MSD	Biosolids	133	186 *5+	186 *5+	84
190-26422-2	Biosolids-Duplicate	150	187 *5+	156 *5+	71
LCS 320-509488/2-A	Lab Control Sample	63	66	76	92
MB 320-509488/1-A	Method Blank	72	77	84	95

Surrogate Legend

PFOSA = 13C8 FOSA
HFPODA = 13C3 HFPO-DA
PFBA = 13C4 PFBA
C3PFBS = 13C3 PFBS
PFDA = 13C2 PFDA
PFDoA = 13C2 PFDoA
C4PFHA = 13C4 PFHpA
PFHxA = 13C2 PFHxA
PFNA = 13C5 PFNA
PFOA = 13C4 PFOA
PFOS = 13C4 PFOS
PFPeA = 13C5 PFPeA
PFTDA = 13C2 PFTeDA
PFUnA = 13C2 PFUnA
d5NEFOS = d5-NEtFOSAA
d3NMFOS = d3-NMeFOSAA
M242FTS = M2-4:2 FTS
M262FTS = M2-6:2 FTS
M282FTS = M2-8:2 FTS
PFHxS = 18O2 PFHxS

Isotope Dilution Summary

Client: City of Cadillac Utilities

Job ID: 190-26422-1

Project/Site: City of Cadillac Biosolids PFAS

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOSA (25-150)	HFPODA (25-150)	PFBA (25-150)	C3PFBS (25-150)	PFDA (25-150)	PFDoA (25-150)	C4PFHA (25-150)	PFHxA (25-150)
190-26422-3	Equipment Blank	87	90	93	89	88	82	94	94
LCS 320-510162/2-A	Lab Control Sample	88	86	88	82	85	82	92	88
LCSD 320-510162/3-A	Lab Control Sample Dup	89	85	87	79	80	87	90	85
MB 320-510162/1-A	Method Blank	90	83	87	80	84	79	87	86

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFNA (25-150)	PFOA (25-150)	PFOS (25-150)	PFPeA (25-150)	PFTDA (25-150)	PFUnA (25-150)	d5NEFOS (25-150)	d3NMFOS (25-150)
190-26422-3	Equipment Blank	101	94	95	92	74	91	89	87
LCS 320-510162/2-A	Lab Control Sample	91	89	92	86	80	86	85	83
LCSD 320-510162/3-A	Lab Control Sample Dup	88	91	91	88	78	92	83	87
MB 320-510162/1-A	Method Blank	86	91	89	86	77	94	86	85

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M242FTS (25-150)	M262FTS (25-150)	M282FTS (25-150)	PFHxS (25-150)
190-26422-3	Equipment Blank	127	140	131	97
LCS 320-510162/2-A	Lab Control Sample	106	116	120	93
LCSD 320-510162/3-A	Lab Control Sample Dup	111	111	102	91
MB 320-510162/1-A	Method Blank	112	117	108	86

Surrogate Legend

PFOSA = 13C8 FOSA

HFPODA = 13C3 HFPO-DA

PFBA = 13C4 PFBA

C3PFBS = 13C3 PFBS

PFDA = 13C2 PFDA

PFDoA = 13C2 PFDoA

C4PFHA = 13C4 PFHpA

PFHxA = 13C2 PFHxA

PFNA = 13C5 PFNA

PFOA = 13C4 PFOA

PFOS = 13C4 PFOS

PFPeA = 13C5 PFPeA

PFTDA = 13C2 PFTeDA

PFUnA = 13C2 PFUnA

d5NEFOS = d5-NEtFOSAA

d3NMFOS = d3-NMeFOSAA

M242FTS = M2-4:2 FTS

M262FTS = M2-6:2 FTS

M282FTS = M2-8:2 FTS

PFHxS = 18O2 PFHxS

Definitions/Glossary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Qualifiers

LCMS

Qualifier	Qualifier Description
*5-	Isotope dilution analyte is outside acceptance limits, low biased.
*5+	Isotope dilution analyte is outside acceptance limits, high biased.
F1	MS and/or MSD recovery exceeds control limits.
I	Value is EMPC (estimated maximum possible concentration).

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
□	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CFU	Colony Forming Unit
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MCL	EPA recommended "Maximum Contaminant Level"
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
MPN	Most Probable Number
MQL	Method Quantitation Limit
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
NEG	Negative / Absent
POS	Positive / Present
PQL	Practical Quantitation Limit
PRES	Presumptive
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)
TNTC	Too Numerous To Count

QC Association Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

LCMS

Prep Batch: 509488

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-26422-1	Biosolids	Total/NA	Solid	SHAKE	
190-26422-2	Biosolids-Duplicate	Total/NA	Solid	SHAKE	
MB 320-509488/1-A	Method Blank	Total/NA	Solid	SHAKE	
LCS 320-509488/2-A	Lab Control Sample	Total/NA	Solid	SHAKE	
190-26422-1 MS	Biosolids	Total/NA	Solid	SHAKE	
190-26422-1 MSD	Biosolids	Total/NA	Solid	SHAKE	

Analysis Batch: 510044

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-26422-1	Biosolids	Total/NA	Solid	537 (modified)	509488
190-26422-2	Biosolids-Duplicate	Total/NA	Solid	537 (modified)	509488
MB 320-509488/1-A	Method Blank	Total/NA	Solid	537 (modified)	509488
LCS 320-509488/2-A	Lab Control Sample	Total/NA	Solid	537 (modified)	509488
190-26422-1 MS	Biosolids	Total/NA	Solid	537 (modified)	509488
190-26422-1 MSD	Biosolids	Total/NA	Solid	537 (modified)	509488

Prep Batch: 510162

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-26422-3	Equipment Blank	Total/NA	Water	3535	
MB 320-510162/1-A	Method Blank	Total/NA	Water	3535	
LCS 320-510162/2-A	Lab Control Sample	Total/NA	Water	3535	
LCSD 320-510162/3-A	Lab Control Sample Dup	Total/NA	Water	3535	

Analysis Batch: 510387

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-26422-3	Equipment Blank	Total/NA	Water	537 (modified)	510162
MB 320-510162/1-A	Method Blank	Total/NA	Water	537 (modified)	510162
LCS 320-510162/2-A	Lab Control Sample	Total/NA	Water	537 (modified)	510162
LCSD 320-510162/3-A	Lab Control Sample Dup	Total/NA	Water	537 (modified)	510162

General Chemistry

Analysis Batch: 509561

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-26422-1	Biosolids	Total/NA	Solid	D 2216	
190-26422-2	Biosolids-Duplicate	Total/NA	Solid	D 2216	

Lab Chronicle

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Client Sample ID: Biosolids

Date Collected: 07/20/21 13:30

Date Received: 07/21/21 10:46

Lab Sample ID: 190-26422-1

Matrix: Solid

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Analysis	D 2216		1	509561	07/23/21 10:03	TCS	TAL SAC

Client Sample ID: Biosolids

Date Collected: 07/20/21 13:30

Date Received: 07/21/21 10:46

Lab Sample ID: 190-26422-1

Matrix: Solid

Percent Solids: 3.5

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	SHAKE			509488	07/23/21 04:42	HK	TAL SAC
Total/NA	Analysis	537 (modified)		1	510044	07/24/21 19:01	RS1	TAL SAC

Client Sample ID: Biosolids-Duplicate

Date Collected: 07/20/21 13:30

Date Received: 07/21/21 10:46

Lab Sample ID: 190-26422-2

Matrix: Solid

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Analysis	D 2216		1	509561	07/23/21 10:03	TCS	TAL SAC

Client Sample ID: Biosolids-Duplicate

Date Collected: 07/20/21 13:30

Date Received: 07/21/21 10:46

Lab Sample ID: 190-26422-2

Matrix: Solid

Percent Solids: 3.6

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	SHAKE			509488	07/23/21 04:42	HK	TAL SAC
Total/NA	Analysis	537 (modified)		1	510044	07/24/21 19:29	RS1	TAL SAC

Client Sample ID: Equipment Blank

Date Collected: 07/20/21 13:15

Date Received: 07/21/21 10:46

Lab Sample ID: 190-26422-3

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			510162	07/25/21 19:29	AP	TAL SAC
Total/NA	Analysis	537 (modified)		1	510387	07/26/21 20:47	S1M	TAL SAC

Laboratory References:

TAL SAC = Eurofins TestAmerica, Sacramento, 880 Riverside Parkway, West Sacramento, CA 95605, TEL (916)373-5600

Analyst References:

Lab: TAL SAC

Batch Type: Prep

AP = Austin Phillips

HK = Harmandeep Kaur

Batch Type: Analysis

RS1 = Rungtip Sanjumnai

S1M = Sudarat Mongkol

TCS = Tammy Saechao

Eurofins TestAmerica, Michigan

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Biosolids PFAS

Job ID: 190-26422-1

Laboratory: Eurofins TestAmerica, Sacramento

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	Identification Number	Expiration Date
Alaska (UST)	State	17-020	02-20-24
ANAB	Dept. of Defense ELAP	L2468	01-20-24
ANAB	Dept. of Energy	L2468.01	01-20-24
ANAB	ISO/IEC 17025	L2468	01-20-24
Arizona	State	AZ0708	08-11-21
California	State	2897	01-31-22
Colorado	State	CA0004	08-31-21
Florida	NELAP	E87570	06-30-22
Georgia	State	4040	01-29-22
Hawaii	State	<cert No.>	01-29-22
Illinois	NELAP	200060	03-18-22
Kansas	NELAP	E-10375	10-31-21
Louisiana	NELAP	01944	06-30-22
Maine	State	CA00004	04-14-22
Michigan	State	9947	01-29-22
Nevada	State	CA000442021-2	07-31-21
New Hampshire	NELAP	2997	04-18-22
New Jersey	NELAP	CA005	06-30-22
New York	NELAP	11666	04-01-22
Ohio	State	41252	01-29-22
Oregon	NELAP	4040	01-30-23
Texas	NELAP	T104704399-19-13	05-31-22
US Fish & Wildlife	US Federal Programs	58448	07-31-21
USDA	US Federal Programs	P330-18-00239	07-31-21
Utah	NELAP	CA000442021-12	03-01-22
Virginia	NELAP	460278	03-14-22
Washington	State	C581	05-05-22
West Virginia (DW)	State	9930C	12-31-21
Wisconsin	State	998204680	08-31-21
Wyoming	State Program	8TMS-L	01-28-19 *

* Accreditation/Certification renewal pending - accreditation/certification considered valid.

Eurofins TestAmerica, Michigan

Ver: 11/01/2020



Environment Testing
TestAmerica

☐ SDS or Known Hazard Information Supplied by Client

☐ Discrepancies

☐ Short Hold

☐ Rush ☐ 24 Hr ☐ 2-Day ☐ 3-Day ☐ 5-Day ☐ Other: _____

Receipt Evaluation Performed by: Initials: TEH Date: 7/21/21 Time: 1040

Client ID: City of Cadillac

Work Order #: _____

Cooler / Sample Receipt

After hours receipt: complete gray

areas. Place cooler in walk-in, place

form in Receiving box. Date: _____ Time: _____

Method of Shipment:

Walk-In Client Eurofins TA Field/Courier

Other Client / 3rd Party Courier: _____

Fed Ex Tracking #: _____

UPS Tracking #: Ground

Other: _____

Shipping Container Type:

☒ Cooler ☐ Box

☐ None ☐ Other: _____

Packing Materials:

☒ Plastic Bags ☐ Foam

☐ Bubble Wrap ☐ Paper

☐ Packing Peanuts ☐ None

☐ Other: _____

Custody Seals Intact:

☒ Yes ☐ No

☐ NA (not used or required)

Cooling Materials:

☐ Ice (Solid) ☒ Ice (Melted)

☐ Blue Ice ☐ None

☐ Other: _____

Bacteriological Samples	Temp Corrected (°C)	Frozen?	Rec'd Within 2 Hrs?	Sample Flagged?
		Yes No	Yes No	Yes No

Received on same day sampled? Yes No

Additional Sheets Required? Yes No

Receipt Temperatures

Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Acceptable	Cooler ID	Affected Samples
<u>CP313207</u>	<u>4.7</u>	<u>4.7</u>		<u>✓</u>	<u>Y</u> <u>N</u>		
					<u>Y</u> <u>N</u>		
					<u>Y</u> <u>N</u>		

Receipt Questions**	Y	N	NA	"No" answers require additional comment
CoC present and ETA receipt signature, date, and time properly documented?	<u>✓</u>			
Containers and Labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)	<u>✓</u>			
Appropriate containers used and adequate volume provided?	<u>✓</u>			Preserved bottles checked for pH?* Yes No
Number of sample containers match CoC?	<u>✓</u>			pH strip lot # _____
Samples received within hold?	<u>✓</u>			
Samples submitted for GRO and Volatiles analysis (8260, 624, 524) received without headspace?			<u>✓</u>	
Was a Trip Blank received with VOA samples?				
Were the samples free of any questionable physical conformities? (i.e.; field duplicates or multiple bottles of the same sample do not significantly vary in appearance – color, solid proportions, etc.)	<u>✓</u>			
Were the CoC bottle labels and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	<u>✓</u>			
**May not be applicable if samples are not for compliance testing				*Excludes FOG, VOAs, TOC Vials, HEM

Client Contact Record

Contact Via: ☐ Phone ☐ Email ☐ Other: _____ Person Contacted: _____ Date/Time: _____

☐ Discrepancy allowance agreement is on record in the client project file

Discussion / Resolution

Any additional documentation and clarification from the client must be noted in the narrative and/or scanned into the CoC directory.

Reviewed by L. Hill Date: 7/21/21

WI-MI-010_020720

